

JARTAI, China

WMO: 535020

Lat: 39.7777N

Lon: 105.7499E

Elev: 3389

StdP: 12.98

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.4	(c) -0.3	(d) -23.3	(e) 1.7	(f) 10.7	(g) -19.4	(h) 2.1	(i) 11.5	(j) 20.1	(k) 25.5	(l) 18.0	(m) 26.9	(n) 4.1	(o) 200	(p) 0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	96.5	63.6	94.2	62.6	91.7	61.8	69.4	85.9	67.6	83.4	66.0	81.9	8.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	103.9	75.5	62.6	96.5	74.4	60.8	90.2	73.6	35.8	85.8	34.2	83.5	32.8	81.8	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	18.7	16.4	-10.2	101.8	5.0	2.5	-13.8	103.6	-16.7	105.0	-19.5	106.4	-23.2	108.2
			-11.1	71.1	4.8	2.1	-14.5	72.6	-17.3	73.9	-20.0	75.1	-23.5	76.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	16.4	25.0	39.5	54.7	66.0	75.4	79.7	76.2	65.8	51.2	34.7	20.1		
	DBStd	23.06	7.50	8.81	9.26	8.61	7.00	5.35	4.90	5.32	6.41	7.57	8.06	7.64		
	HDD50	3598	1040	699	341	51	2	0	0	0	0	78	459	928		
	HDD65	6610	1505	1119	791	320	72	3	0	2	67	430	908	1393		
	CDD50	3790	0	0	15	193	499	761	921	811	474	115	1	0		
	CDD65	1328	0	0	0	12	105	314	456	348	91	2	0	0		
	CDH74	15511	0	0	5	236	1399	3786	5496	3706	842	41	0	0		
	CDH80	7105	0	0	0	61	484	1780	2868	1667	242	3	0	0		
(14) Wind	WSAvg	7.1	6.1	6.7	7.8	8.5	8.1	7.5	7.2	7.1	6.4	6.4	6.9	6.6		
(15) Precipitation	PrecAvg	4.1	0.0	0.0	0.1	0.2	0.3	0.5	1.0	1.2	0.5	0.2	0.1	0.0		
	PrecMax	8.9	0.2	0.3	0.4	0.8	2.1	1.7	3.7	4.9	2.5	0.8	0.8	0.2		
	PrecMin	1.9	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.1	0.0	0.1	0.2	0.4	0.4	0.9	1.0	0.6	0.2	0.2	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	54.3	72.0	86.9	92.3	97.8	101.4	97.7	92.2	78.9	62.2	45.3		
		MCWB	30.6	37.2	46.8	54.0	56.8	61.3	66.4	64.9	59.8	53.4	42.5	32.5		
	2%	DB	37.4	48.6	65.4	80.6	88.6	94.8	97.7	94.3	85.7	73.7	57.2	40.2		
		MCWB	27.3	34.5	43.2	50.7	55.9	60.2	64.7	64.1	58.5	50.7	40.0	29.4		
	5%	DB	33.3	44.6	60.7	75.6	85.4	92.2	95.2	91.6	82.1	69.2	52.6	36.1		
		MCWB	24.8	32.0	40.9	48.8	54.5	59.6	63.9	63.2	57.4	48.8	38.1	27.1		
	10%	DB	29.7	40.7	56.0	71.4	82.0	89.5	92.6	88.9	78.8	65.2	48.3	32.8		
		MCWB	22.4	29.8	38.3	47.1	53.1	58.9	63.5	62.4	56.9	47.1	35.9	25.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.9	39.1	47.9	56.3	60.8	66.9	72.5	72.2	65.8	56.5	44.0	32.7		
		MCDB	42.4	48.5	68.2	77.7	79.5	81.7	93.9	86.3	78.0	69.8	58.1	44.5		
	2%	WB	27.5	35.1	44.1	53.2	58.4	64.8	69.8	69.7	63.3	53.2	41.1	29.7		
		MCDB	36.9	47.7	63.0	73.0	80.8	80.1	88.4	84.2	76.3	68.5	54.0	39.0		
	5%	WB	25.0	32.5	41.7	50.6	56.5	63.3	68.0	67.7	61.3	50.7	38.9	27.3		
		MCDB	32.9	43.6	58.5	72.6	79.2	79.9	84.8	82.4	74.3	65.2	51.3	35.7		
	10%	WB	22.5	30.1	39.3	48.2	54.8	61.9	66.6	65.9	59.1	48.5	36.6	25.1		
		MCDB	29.5	40.6	54.0	69.6	77.5	80.2	82.9	81.5	73.2	62.7	47.7	32.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	23.5	25.5	25.5	25.0	24.4	22.8	21.4	21.1	21.5	23.1	21.8	21.1		
		MCDBR	26.7	30.0	31.5	31.4	29.8	27.4	25.7	25.0	26.5	28.3	25.8	24.0		
		MCWBR	19.1	19.2	16.8	14.4	11.8	8.5	7.2	7.2	9.0	12.6	14.3	16.5		
	5% WB	MCDBR	26.3	28.4	28.4	27.2	24.5	20.7	20.7	19.3	19.4	24.6	23.7	23.0		
		MCWBR	18.9	18.3	15.3	12.8	10.3	6.7	6.3	6.9	7.8	12.0	13.5	16.0		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.382	0.431	0.478	0.502	0.516	0.499	0.490	0.464	0.411	0.353	0.324		
	taud		2.102	1.976	1.867	1.795	1.803	1.834	1.949	1.952	1.946	1.997	2.081	2.102		
	Ebn at Noon		255	256	256	253	250	246	249	247	244	242	243	246		
	Edh at Noon		35	47	58	67	68	66	58	56	53	45	36	33		
(44) All-Sky Solar Radiation	RadAvg		856	1177	1577	1940	2180	2141	2010	1865	1564	1302	954	778		
	RadStd		75	63	97	105	85	149	103	81	111	56	63	55		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page